

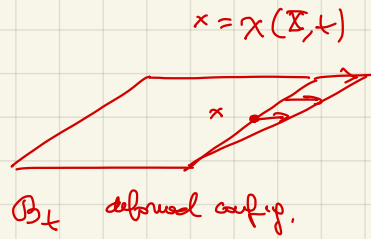
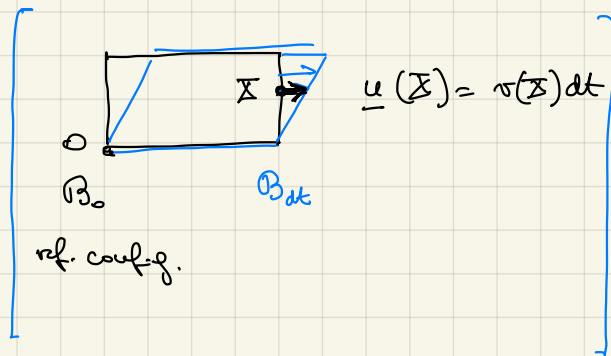
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14/04/26

Linear Elasticity

Start a "flow" from a ref. config. (typically, the natural, unstressed config. of a material)



This is the domain of linear elasticity

• Translations: $\underline{u}(\underline{X}) \equiv \underline{u}_0$

• Rotations: $\nabla \underline{u}(\underline{X}) \equiv \underline{W}$ const, $\underline{W} \in \text{skw}$

$$\begin{bmatrix} 0 & \gamma & -\beta \\ -\gamma & 0 & \alpha \\ \beta & -\alpha & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & -\gamma & \beta \\ \gamma & 0 & -\alpha \\ -\beta & \alpha & 0 \end{bmatrix}$$

Re: $(\nabla \underline{u})_{ij} = \frac{\partial u_i}{\partial x_j}$

3x3 in 3D

There exists $\underline{\omega} = (\alpha, \beta, \gamma) \in \mathbb{R}^3$ s.t. $\underline{W}(\underline{X}-\underline{0}) = \underline{\omega} \wedge (\underline{X}-\underline{0})$

Check:

$$\begin{bmatrix} 0 & -\gamma & \beta \\ \gamma & 0 & -\alpha \\ -\beta & \alpha & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\gamma x_2 + \beta x_3 \\ \gamma x_1 - \alpha x_3 \\ -\beta x_1 + \alpha x_2 \end{bmatrix}$$

$$\begin{bmatrix} e_1 & e_2 & e_3 \\ \alpha & \beta & \gamma \\ x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} \beta x_3 - \gamma x_2 \\ -\alpha x_3 + \gamma x_1 \\ \alpha x_2 - \beta x_1 \end{bmatrix}$$

$\underline{\omega}$ is called the axial vector of $\underline{W}$; $\underline{\omega}$ is called the axial vector of $\underline{W} \in \mathbb{R}^3$

$\in \mathbb{R}^3$ $\in \text{skw}$ $\in \text{skw}$

$\underline{v}(\underline{X}) = \underline{\omega} \wedge (\underline{X} - \underline{0})$

$\underline{u}(\underline{X}) = \underline{v}(\underline{X}) dt = \underline{\omega} dt \wedge (\underline{X} - \underline{0})$

• Linear or infinitesimal theory $\underline{E} = \text{sym } \nabla \underline{u} = \frac{1}{2} (\nabla \underline{u} + \nabla \underline{u}^T)$

$$E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

Affine deformations: $\underline{u}(\underline{x}) = \underline{u}(0) + \underline{A}(\underline{x} - 0)$

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$$\nabla \underline{u}(\underline{x}) = \underline{A} = \underbrace{\text{sym } \underline{A}}_{\underline{E} \in \text{Sym}} + \underbrace{\text{skw } \underline{A}}_{\underline{W} \in \text{Skw}}$$

$\underline{E} \in \text{Sym}$ Linear or infinitesimal strain
 $\underline{W} \in \text{Skw}$ $\underline{W} dt$, axial vector of $\underline{W} \in \text{Skw}$, infinitesimal rotation

$$\underline{u}(\underline{x}) = \underline{u}(0) + \underbrace{\underline{W}(\underline{x} - 0)}_{\text{rotation}} + \underline{E}(\underline{x} - 0)$$

$$= \underline{W} dt \wedge (\underline{x} - 0)$$

= translation + infinitesimal rotation + infinitesimal strain

- Remark 1: general deformations are non-affine

but by Taylor expansion at order 1 in $|\underline{x}|$ near an arbitrary point $\underline{0}$, we can analyze deformations in a neighborhood of $\underline{0}$ as affine deformations

- Remark 2: in nonlinear kinematics, the decomposition is

translation "+" rotation "+" strain
 is NOT additive but multiplicative.

The "+" is actually a composition of maps.

$$\text{translation}(\text{rotation}(\text{strain}))$$

For example: $\text{rotation}_1(\text{rotation}_2)$

their gradients are rotations R_1, R_2 and by chain rule

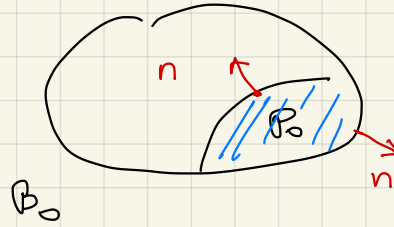
$$\nabla(\text{rotation}_1(\text{rotation}_2)) = R_1 R_2$$

Rotations are multiplied, not added (unless they have the same axis)

Stresses and balance (equilibrium)

$$\operatorname{div} \underline{T} + \underline{b}_0 = 0$$

$$\int_{P_0} b_0 dV + \int_{\partial P_0} \underline{T} \underline{n} dA = 0$$



Constitutive equations

$$\underline{T} = 2\mu \underline{E} - p \underline{I}, \quad \operatorname{div} \underline{u} = 0 \quad (\text{incompressible, isotropic solid})$$

$$\rightarrow \underline{T} = 2\mu \underline{E} + \lambda (\operatorname{tr} \underline{E}) \underline{I} \quad (\text{compressible, isotropic}) \leftarrow$$

$$\underline{T} = \underline{C} \underline{E}$$

$$T_{ij} = \sum_{k,l=1}^3 C_{ijkl} E_{kl}$$

(general Hooke's law, $\underline{C}$ 4th order tensor of elastic moduli / stiffnesses)

$$\underline{T} = 2\mu \underline{E} + \lambda (\text{tr} \underline{E}) \underline{I} \quad (.)$$

Goal: find inverse relation, i.e., $\underline{E}$ as a function of $\underline{T}$

$$\text{To show: } \underline{E} = \frac{1}{2\mu} \left[\underline{T} - \frac{\lambda}{2\mu+3\lambda} (\text{tr} \underline{T}) \underline{I} \right] = \underline{C}^{-1} \underline{T} \quad (..) \\ \downarrow \text{compliance matrix}$$

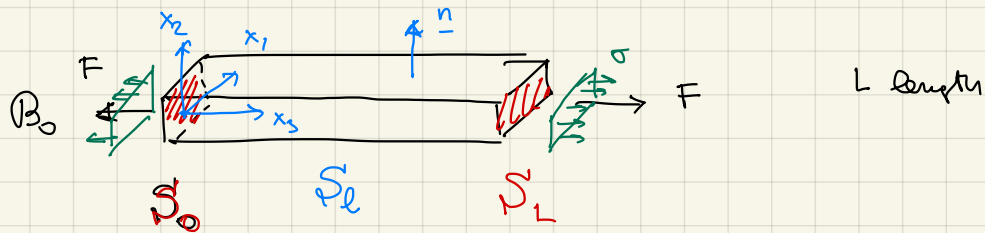
Take trace of (.)

$$\text{tr} \underline{T} = 2\mu \text{tr} \underline{E} + \lambda (\text{tr} \underline{E}) 3 = (2\mu + 3\lambda) \text{tr} \underline{E}$$

$$\text{tr} \underline{E} = \frac{1}{2\mu + 3\lambda} \text{tr} \underline{T}$$

$$\underline{E} = \frac{1}{2\mu} \underline{T} - \frac{\lambda}{2\mu} (\text{tr} \underline{E}) \underline{I} = \frac{1}{2\mu} \underline{T} - \frac{\lambda}{2\mu(2\mu+3\lambda)} (\text{tr} \underline{T}) \underline{I} \quad \text{which is } (..)$$

Some simple solutions: uniaxial extension, no body forces.



$$\underline{T} \underline{e}_3 \Big|_{x_3=L} = \sigma \underline{e}_3 ; \quad \underline{T} (-\underline{e}_3) \Big|_{x_3=0} = -\sigma \underline{e}_3 ; \quad \underline{T}(\underline{n}) \Big|_{S_L} = 0 \quad (\text{BC})$$

$$d\mathbf{T} = d\mathbf{T}(\mathbb{C}(\text{sym} \nabla \underline{u})) = 0$$

$$\underline{T} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma \end{bmatrix} \text{ satisfies both } d\mathbf{T} = 0 \text{ (trivially because of BC's)} \text{ (also constant)}$$

$$\underline{E} = \frac{1}{2\mu} \left[\underline{T} - \frac{\lambda}{2\mu+3\lambda} (\text{tr} \underline{T}) \underline{I} \right]$$

All off-diagonal terms vanish

$$\underline{E}_{11} = \underline{E}_{22} = \frac{1}{2\mu} \left[0 - \frac{\lambda}{2\mu+3\lambda} \sigma \cdot 1 \right] = - \frac{1}{2\mu} \frac{\lambda}{2\mu+3\lambda} \sigma$$

$$\underline{E}_{33} = \frac{1}{2\mu} \left[\sigma - \frac{\lambda}{2\mu+3\lambda} \sigma \cdot 1 \right] = \frac{1}{2\mu} \frac{2\mu+3\lambda - \lambda}{2\mu+3\lambda} \sigma = \frac{\mu+\lambda}{\mu(2\mu+3\lambda)} \sigma =: \frac{\sigma}{E_y}$$

In this way, $E_y = \frac{\sigma}{E_{33}} \Rightarrow E_y = \frac{\mu(2\mu+3\lambda)}{\mu+\lambda}$ Young's modulus

$$\underline{E}_{11} = \underline{E}_{22} = - \frac{\lambda}{2} \frac{1}{\mu+\lambda} \frac{\mu+\lambda}{\mu(2\mu+3\lambda)} \sigma = - \nu \underline{E}_{33} \quad \text{where } \nu = \frac{\lambda}{2(\mu+\lambda)} \text{ is Poisson's ratio}$$

$$= - \frac{\underline{E}_{11}}{\underline{E}_{33}} \text{ ratio of lateral contraction}$$

Thermodynamics dictates $\nu \in (-1, 1/2)$